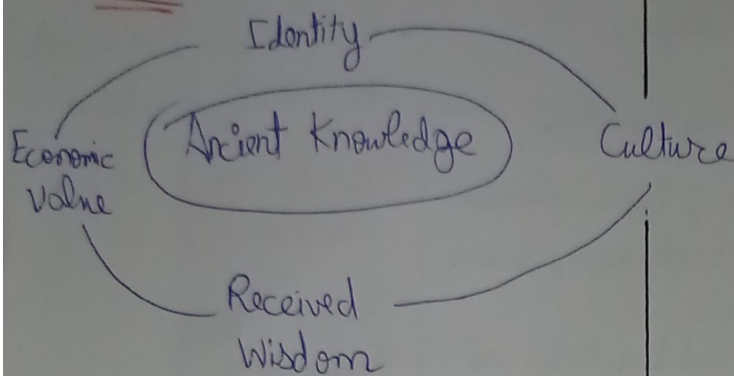


IKS



- Creative Phase
- 5000 years old recorded
- Unrecorded
- Vedic Period
- Wisdom of Past

Imp of ancient knowledge

- Decimal Sys.
- Old civilization
- Cultures
- Steel,
- Contribution (16 pts)

Sanatana Dharma

- Hinduism
- Eternal & absolute set of duty
- Gradualists
- Swadharma - One's own duty
- Bhagavata Crita
- 4 Veda → ~~Attharvaveda, Yajurveda, Rigveda, Samaveda~~
- Conflict b/w Swadharma & Sanatana Contrast
- Puranas, Upanishad & Shashtra

Puranas

- Vast genre of Indian Lit
- Narratives of history
- Cosmology & philosophy
- Vishnu Purana (4 ved)
- Shukla
- Skanda
- (Ramayana, Mahabharata)
- Valmiki
- Ved Vyas

Why

- Thinking Pattern
- Ancient knowledge analysis
- Innovation
- Scope for research
- Rich heritage knowledge
- Spiritual preserved

Quote

- We owe a lot to the Indians, who taught us how to count, without which no modern scientific discovery could have been made.

- Albert Einstein

Current Status

- Not good
- Digging
- Neglecting knowledge
- Hub of knowledge :- Vishvaguru
- Lost inner Peace

Chaturvedsha Vidyasthana

- ↳ 14 branches of learning
- (Foundation) 4 Ved ⇒ Rig, Yajur, Sama, Atharva
- 6 Vedanga ⇒ Siksha, Kalpa, Vyakarna, Nirukta, Chandas, Jyotisha
- (Right Passage) (Phonetics) (Grammar) (Rhetoric) (Metrical Science) (Astronomy)
- 4 Upadev ⇒ Arthashastra, Grandhara, Ayurved, Dhanurved

Niti Shashtra

- Moral values
- Behaviour in society
- Chanakya
- Truth is bitter of all

Itihās

Dharma, Artha, Kama, Moksha

Samanya

- Common Sense
- Self Respect
- Moral

Diff b/w Indian & West

- I
- Rich body of knowledge
- Inherit from past
- Spiritual Beliefs
- Collectiveness
- W
- Physical, Social
- Individual

Veda (Apauruseya)

- Not the work of only men
- Not written by Seers
- If So - Mantra Books
- Aryan dynasty

Subdivision of Vedas

- Samhitas (Mantras)
- Aranyakas (Rituals ceremony)
- Brahmanas (Commentaries)
- Upanishads (Meditation etc.)
- Meditation

Salient Features of Vedas

- Panchtatva Worship
- Social Classes
- Aay Iron Tools
- First development
- Crafts

Buddha School

- Gautam Buddha Teaching
- 4-5 BC
- Ancient India
- Spread across Asia
- Northern, Southern (China) (Africa)
- Life and death suffering

Teachings :- Suffering,
Four elements, Consciousness,
To reach Nirvana

Carvaka School

- Non vedic
- Realistic beliefs
- No space
- Matter is the ultimate reality
- World is made of 4 basic elements
- Goal of life is to experience pleasure 'here and now'.
- Consciousness and knowledge are embodied, not eternal
- No afterlife, rebirth
- No room for any of transcendental entities

Ramayana

- Idealism
- Non realistic

Mahabharata

- Real life learning
- Realistic
- Mistakes have been mentioned

Jaina School

- Sacrifices
- 24 Tirthankaras
- 24th Mahavir
- endless birth-death
- Consider tiny particles
- Ahimsa
- Karma

Key of Mahabharat

- Justice prevails
- Fight for our rights
- Revengeful instinct can only lead to one's doom.
- Stand by what's right
- Half knowledge is dangerous
- Don't be too swayed by greed
- Being a woman does not make you unless a individual

Components of Language

- Vedanga
- Linguistics → Pronunciation, speech sound, structures

Panini's Work on Sanskrit Grammar

- Study of Adhyay + Linguistics
- Phonetics
- Grammar
- Best descriptive model of Grammar
- 3983 rules in Sanskrit language
- Distinguish features make Sanskrit powerful language.

Number System

- 300 BCE
- Decimal Sys
- Brahmagupta developed symbol

$$\pi = \frac{2827433388233}{9 \times 10^{11}} \\ = 3.14159265359222$$

4 Veda

Rig ⇒ Religious
(Upanishads)
Sām ⇒ Singing
Yajur ⇒ Sacrifices
Atharva ⇒ Magic & Charms
Nam ⇒ Subanta
Verb forms ⇒ Tinanta
Word = Base + Suffix

Maheśvara Sūtra

↓
Alphabets

Karaka $\Rightarrow$ Conjunctions
(Link the work)

Upa-sargas $\Rightarrow$ Prefixes
↓
22

NLP

Branch of linguistics

Processing

14 determiners

Unique Aspects of Indian Astronomy

- Modern Science - Astronomy space, Celestial objects are diff
- No relation with western society
- Great impact on Indian society
- Zodiac sign (Rasi), Nakshatra (Star) Calendar (Panchanga)
- Integral part of living being - Celestial obj
- Determination of Time
- Advance maths based on astronomical obs.

Celestial Coordinate System

- Reference pt. on earth wst which all other planet's pos is identified
- ~~An elliptical path with reference~~
- A sphere ^{was} considered with reference to earth and observer is on earth who observe the motion of sun and from that other ~~information~~ positions were identified
- Sun ~~was~~ tilted at 23.5° around axis
Sun revolve around earth
Zenith - South Pole North Pole
Nadir - South Pole
Azimuth - Equator line
- Angle b/w projected vector & reference vector on the reference plane is called Azimuth
- All planets rise from east like sun. (Celestial obj)
- Fix the pos of stars
- Moving Bodies - Planets
- West to east move - Planets
- ~~moving~~ wst planets pos of stars & sun change, then month, year & days were identified

Historical Development

- Developments based on astronomy
- Navigational purpose
- Seasonal Info
- Sky as abode of gods
- Cardinal dir to set vedic altars
- Helps in Trade & agriculture
- Predict whether condition
- Latitude, Longitude
- Navigation
- Atharvaved - Stars, planets
- Yuga = 5 solar years, 67 lunar sidereal cycles, 1830 days & 1835 sidereal days
- Nilkantha Somayaji was ~~the~~ already given the Kepler model
- Yuga $\rightarrow$ to synchronise solar & lunar ~~month~~ calendar
- Add 2 months to yuga to complete it
- Sidhanta Astronomy $\rightarrow$ Mathematics
Add Planets & info
^{more}
- Nilkantha Calculated planetary motion
↳ Planets orbit around the sun
↳ sun orbits around the earth
- Sun $\rightarrow$ Northern most pt. $\rightarrow$ Summer Solstice
- Sun $\rightarrow$ Southern most pt $\rightarrow$ Winter Solstice
- Zodiac Sign $\leftarrow$ New pos of stars
- Each day $\rightarrow$ Nakshatra
would be associated
- Sidereal $\rightarrow$ Time taken by obj to complete 1 rev in the background of stars.

Elements of Indian Calendar

- Based on sun & moon
- Year = Take a reference pt as a star, sun ~~revolve~~ time taken to reach that reference pt is ~~the~~ year.
- Lunar ~~year~~ = Moon with sun month
- Solar calendar = Dates of year
- Lunar Calendar = Religious festivals, auspicious times.
- ~~Used~~ Luni-Solar calendar is used in India
- Vedic Year = 12 month $\rightarrow$ 30 days
 $\downarrow$
 Total = 360 days
 $\downarrow$
 To synchronise with season $\rightarrow$ add 5
 $\downarrow$
 365 days
- Lunar Calendar = 354 days + 36 days
- Gap = 10.89 days
- To align we add a ^{Lunar} month after few years
- Yajurved $\Rightarrow$ Add 2 months after 5 years
- Pos of sun & moon return together at some pos in beamfork of stars after 5 years.
- Solar Month :- Time taken by sun to ~~revolve~~ ~~to~~ traverse from Rasi to another Rasi
- Lunar Month :- Time b/w 2 new moons (full)
- Tithi :- Relative pos of moon & sun on a day. Diff of 12° in every tithi (15^{th} day = 180° full moon)

Nityabhata & Siddhantic Tradition

- Sophisticated & elaborated
- Nityabhata was pioneer
- Spread full-fledged mathematical astronomy
- Added more planets
- Nityabhata - Pos of sun, moon & planets. Astronomical variable.
- 121 verses organised by 4 Pada
 Critikapada, Granitapada, Kalakriyapada, Gulasopada
- "The globe of the earth stands at the centre of the circular prom of asterisms surrounded by the earth orbits, it is made up of water, earth, fire & air & is spherical".
- Acc to Nityabhata only earth is ~~revolve~~ rotating at its axis & other are stationary
- 4 mahayug - Treta, Dvapara, Kali, Karta
- Earlier ratio - 4:3:2:1
 New - Yuga duration - 10,80,000 ^{yes.}
- First framework of mathematical astronomy in India.
- Apparent motion of sun, moon & planets are not uniform in background of stars.
- 2 corrections - Manda-Samskara, Sighra-Samskara

Panchanga

- Backbone of Indian astrology in calendar
- Festivals & local events are based on Panchanga
- Phys vital role in daily life
- Can be traced back to Vedanga jyotisa
- 5 years = 1 Yuga
- $\frac{1}{2}$ Tithi = Karana = 6° diff
- ~~At present~~ $\rightarrow$ current = 57th
- Nakshatra $\rightarrow$ In ~~the~~ ecliptic $\rightarrow$ 27
 $\uparrow$
 equal part
- Vra $\rightarrow$ Days of week = 7 days

Jantar Mantar of Raja Sai Sawai

- Raja of Jaipur
- Astronomer - Training in early life
- Collected text of Indian, European, & arabic tradition & studies them himself
- He didn't like use of old data in Indian astronomical constants, so he prepared massive instruments that were ~~study~~ sturdy free from wear-tear or affected by whether condⁿ.
- These observatories were known as Jantar Mantara.
- There are 5 in Delhi, Jaipur, Varanasi, Ujjain & Mathura.
- Mathura $\rightarrow$ destroyed due to attacks

Astronomical Instruments

(Yantra)

- We observe movements & pos of heavenly bodies before a theory to explain their motion can be propounded.
- If prediction does not tally with obs $\rightarrow$ theories have to be revised.
- Made a variety of instruments bcz of inaccurate obs.
- Water clock - Ghati - Yantra
- Sun dial - Kapala Yantra
- Gnomon

Unique Aspects of Indian Mathematics

- Maths blend of poetry, lit, logic & mathematical thinking
- Use of Sutras & pithy verses
- Divided into two based on liking of maths bcz left side of brain is of maths & other for lit.
- Indian maths is mix of both so no stress of learning maths
- Considered as a part of life & inscriptions were found in temples. Ex - Bhaskaracharya & Shankaracharya
- $\downarrow$
Philosophical issue
example were used
to explain decimal sys.

$\downarrow$
Gave riddle to
students
- Aryabhata wrote Pingala where binary cycle was explained
- Aryabhata sys of numeration is an easy way to learn sine table.
- Different mathematical thinking were developed
- Indian mathematics is based on algorithmic approach & that it allows for approx sol. based on needs of real life sit.

Greatest Mathematicians & their Contributions

- Deal with almost all areas of modern maths for more than 1000 years
- Buddhist & Jain works suggest the maths culture & concepts that were thousands years old

Vedic Texts - 3000 BCE or earlier
Num sys, Pythagorean triplet, decimal sys,
Concept of ∞

Panini - 500 BCE

Algorithmic approach, BNF, Prog lang,
Arrays, inheritance, polymorphism

Buddha - 500 BCE to 500 CE

Multivalued logic, interminate & ∞ num

Aryabhata - 476 - 550 CE

Concise verses, sq root, cube root, value
Sys, Sine table, Geometry, quad eqn, Sums
of sq & cubes, Planetary astronomy, plane
& spherical trig

Vishnukata - 600 CE

Fibonacci Numbers, Magic metres

Jyadva - 10th cent CE

Method of soln of 2nd order interminate
eqn

Shripati - 1019-1068 CE

Planetary Astronomy

Trigo (Jyotipatti)

- Aryabhata discovered 2 methods for
Sine table - Geometrical & analytical

Arithmetic

- Branch of maths dealing with computation
using num
- Part of daily life
- Basic ops - $+$, $-$, $\times$, $/$
- Sophisticated at the time of Aryabhata
due to decimal place value sys.
- Indians possess a method of calc that
no word can praise enough. Their rational
sys of maths or their method of calc.
 $\Rightarrow$ I mean the sys of using nine symbols
- Sq of Num - Algo first by Aryabhata
- Shukla gives a good desc of calc with
 $+$, $-$, $\times$, $/$, num, zero, frac & surds.
- Varga is a sq (acc of Aryabhata)

Geometry

- Ved $\Rightarrow$ Vajna is a imp part of life. The performance of
vedic rituals involve conc of vedic alters
as per certain specifications.
- Sulba - Sutra is a part of vedanga that gives
info on methods of layout of vedi, siti &
mandapa
- Sulba $\Rightarrow$ Thread
- π $\Rightarrow$ Earliest Indian works take value of
 π as 3. Sulba - Sutra estimate it to 3.088
- ~~Shukla~~
Bhaskaracharya & Aryabhata estimate 3.14
by their own methods.

Algebra

Analysis - Bija Granit

- Become core of maths in 20th cent which many
historians termed as "algebraization of mathematics".
- Divided into two parts one with analysis & other
with that are essential for analysis
- Deal with signs, arithmetic of 0, & with unknown
- Kuttaka - linear interminate eqn

- Yu - Yuta addition
- Ksa - Kasya subtraction
- Gru - Grana - multiplication
- bha - Bhaga - division
- mu or ka - Mula or Kotani root
- Indians analysed & classified eqnⁿ as - Samikarana

Q. Magic Squares

- Sometimes ^{maths} can be a tool for human curiosity, fun & mind expanding exercise - For ex - Magic sq
- An array of numbers containing an equal num of rows & col is called a magic sq
- Summation of num of every row & col are same
- Pan diagonal Magic sq - If summation of other diagonal is also same
- Matrix like is like a torus (rolled up)
- Indian mathematician specialized in one of special class of magic sq called as Pan diagonal magic sq. For ex - Bhadra - Granita

Binary Mathematics

- Work of Pingala - Chandah-Shastra
- The metres in Sanskrit were analyzed by 2 kind of syllables laghu (L) & Guru (G) which basically is 1 & 0 making binary
- These concepts introduced by Pingala around 300 BCE
- Pingala laid a solid foundation for combinatorial theory & binary mathematics which was followed by later-day Mathematicians in the country

Const of 4x4 pan diagonal magic sq

	3		
5	16	2	
	9		
			1

10	3	13	
5	16	2	
4	9	7	
			1

10	3	13	8
5	16	2	
4	9	7	14
15		12	1

10	3	13	8
5	16	2	11
4	9	7	14
15	6	12	1

UNIT-4

Indian S & T Heritage

- Archaeological evidence that shows how S & T ought to have been developed in ancient times
- Living examples of physical entities that pt. to the use of good S & T practices
- Literary resources pointing to the use of S & T in ancient times
- Archaeological excavations in last 150 yrs have fetched the rich collec of artifacts that reveals clues to engineering & technology

- For ex - Excavation in Rajasthan & Gujarat reveal full functioning town planning & dwelling ~~houses~~ houses built of standard prop of dimensions
- Excavation shows Lothal (Gujarat) had a man made dock for small & big boats
- Several travellers accounts reveal how famous Indian metal smiths were for their quality & workmanship

Mining And Ore Extraction

- Good quality & design of ores & metalworking.
- Unique skills in mining of several metals
- Good design of carving of rocks in many heritage sites
- H Z L research program - Zinc recovery
- Extraction of ore with haph hammering, chiseling & scraping tool
- Udaipur - H Z L (1982)
- Copper Mining - Khetri (Rajasthan)
- Site were remained aside on fire for 3 days and after that remaining work were done (chisel lamps)

Cu Mining - • Utensils, Tools, weapons

- Copper bolt, Buddha Statue (Sultanganj) (Ashok Pillars), 80ft statue in Malanda
- Brass - Cu & Calamine
- Coins were principle use
- 2 variety - Pure & Block Cu
↓
Nepal Cu (99.5%)
- Powder Cu - Therapeutic comb
- Bronze - Cu & Tin (Bells)
↳ Large Amt.
- Bronze Tech - Boffling
- Water & sand clock in Kerala
- ~~Pancha~~ Pancha-loha $\Rightarrow$ Gold, Silver, Cu, Zn & Iron

Lead & Silver

(1000°C)

- Zawar - Roasting & red in open furnaces then reduced & separated

Metal & metalworking technology

- Ornaments, vessels, artifacts, statues, coins, etc. by this we know that Indians were advanced in iron, steel & zinc
- Alloys were discovered
- Brass were found in Mohenjodaro & Harappa
- Bronze statue in Harappa
- Good knowledge of temp & other factors (better than west) (Whit & Steel)
- First to introduce Zn & Cu-Zn alloys
- Amalgam - Gold + Hg (In nature)
- Gold & HgS - Ayurvedic Med
- Hg - Amalgamation Process
- Add water & roll in rag & then Hg is extracted & then we burn
- Convert into sheets & combined with Cow-dung then pure gold is extracted
- Zinc - Last discovered, Rajasthan, 11th cent, 15000 ton from 2.5 million ton ore

Zinc Smelting - Industrial level by making of zig-zag paths for transporting

- Distillation of water, wine & Hg was probably known even in the vedic period. Found in Takshila (Apparatus)
- Dambau Yantra - Apparatus
- Zawar smelters used advanced methods

Iron & Steel In India

- Found in many places
- Christian era
- Qutub Minar - Iron Pillar
- 29 Iron beams - Konark Temple in ~~Odisha~~ Odisha
- Weapons & Tools - Sword, Arrow, Javelin, Daggers
- Surgical 100 tools - ^{In} Sushruta, Samhita (Fe & C alloys)
- Carburization Iron = Wootz steel
 - ↓
 - Damascus Blades
 - Military app

- Bangladesh - Conon (High Quality of Iron & Steel)
- Delhi Iron pillar famous for corrosion resistance (1600 years) bcz of low humidity
- Hematite, Magnetite, limonite - Hydrated Form
- Furnaces - Clay blocks, reusable
- Smelting done by tribes whose god were Ashu
- Kanta Loha (Soft Iron) - Mag
- Tiksa Loha (Carbon Steel) - Hard & Tough
- Munda Loha (Cast Iron) - Brittle & Low MP

Manufacture Steel

- Cast iron - C = Steel
- Cementation (England - 18th cent CE)

Lost wax casting of Idols

- ~~Re~~ Chola era - Icons & Idols using Bronze
- Technique - 6000 year old amulet from Indus - Saraswati civilization
- Beeswax - Remains of beehives
 - ↳ Mold

Apparatus used for extraction

- Engineer & metal smith - 1st grp
- Ayurvedic
- Alchemist - Small scale
 - ↳ Extraction of gold
- Crucibles - Musa - 17
- 51 metallurgical tools - Upkoshana
- 36 Yantra
- 9 Furnaces
- All 3 grps used each other's tool
- Dola Yantra - Uses suspension mechanism (Hung Cured)
- Swedani Yantra - Steaming
- Patana Yantra - Used for sublimation & distillation process
- Dheki Yantra - Distillation of Hg
- Valuka Yantra - Sand bath
- Dhupa Yantra - For heating

Literary Sources of Science

- No corrosion in old iron pillars
- Good knowledge of extraction
- Iron & no copper
- Atharva, Vajurved - Mention of 6 metals
 - ↓
 - Gold, Silver, Iron, Tin, Lead, Cu
- Chandaogya Upanishads
- Ramayan - Iron etc, metals, mine
- Panchashastra - State Crafts & public administration

Alchemy - Transmutation of metals
for gold

Nagarjun - Indian alchemist - Buddhist monk
Text - Rasaratnakara

↓
Apparatus for extraction
process

Town Planning - Great knowledge of
architecture

Medical App - Sushruta-Samhita,
Caraka-Samhita

Irrigation & Water Management

- Agrarian-based economy
- Earlier was very good - Dholavira
↓
Rectangular Step well
3 times bigger
than Mohenjo-daro
- Sonchi & Sudarshana Dam
- Chandragupta Maurya - Sudarshana
Dam in Junagadh
Gujarat
- Maurya's tax - Flood mitigation

Wall Painting

- Adornments were prepared & coated on
the walls
- Buffalo hide - Boil in water - Cut in
small pieces - dried till hot &

Surgical Technique (Salya Tantra)

- Pre Christian era
- Surgery - Heal the wound
- Training of surgeons
- Use of hooks of several trees

↓
Oldest Branch
of Indian medicine

Physical Structures in India

- Old building
- Can suffer from Catastrophic damage
tolerate

South Indian - Chola,

~~Thanjavur~~

Thanjavur - 208 ft high temple

Dyes & Painting Technology

- Adding colours to life both in a figurative
& real sense
- Knowledge of diff colours & techniques of
making dyes → Organic & inorganic
- Wool - Easy
- Cotton - Difficult
- Colouring agents - Pigments
- Mohenjo-daro - Indus Valley knows
dyeing
- Indigo, Turmeric, Resin
- 6 sources of dyes - roots, trunks, barks,
leaves, flowers & fruits
- Palasha Tree - Essence of colours.
- Dyes are also used in cosmetics
- knew the art of selecting colors & mixing
them to make diff shades
- Ajanta paintings employed extracts of
plants & minerals
- Vishnu - dharmottara Purana discusses
diff aspects of painting the thread.

Ship Building

- Maritime Trade
- Tamil Nadu - Raja Raja Chola
- Madurai Kanki, Pura-Nanuru, Aha-nanuru
Songam period texts

- The Jataka stories shows Buddhist were involved in maritime trade

64 Art Forums

- Chatuh-sasti Kala or vidya
- Artistic skills, Beautifying body, Skills of doc, cooking related, entertainment skills, Intellectual pursuits, Engineering & technical skills, Uncommon skills

Status of Indigenous SAT

- Everything is linked to SAT